Order Saturation: deterministic pilot observations

Observation-only status

This report records execution outputs; it makes no support, weakening, priority or scientific-status decision. Independent mathematical review remains separate. All instances are supplied synthetic lattices.

Trial and provenance

Experiment EXP-ENDO-854769; hypothesis H-ENDO-ccf484; run RUN-ENDO-910c24. Code commit 1e4fd5c77cf9f4ef7337ad0a9f29568174538a4b. Seed 1729. Exact command: ["python3", "tools/lattice_order_saturation.py", "--seed", "1729", "--output-dir", "/workspace/lattice-saturation/experiments/EXP-ENDO-854769/runs/RUN-ENDO-910c24"]. One trial requested; completion/failure and resource observations are preserved in execution-receipt.json. The driver records full-tree dirtiness as unknown when not supplied; every source byte is admitted against its committed hash.

Recorded outputs

Exact fixture counters (integer counts, not independent statistical samples): {"all_exact_checks_passed": true, "division_controls": 512, "order_cases": 32, "policy_arms": 64, "wrong_ambient_collisions": 32}. Supervisor receipt: {"artifact_sha256": {"check.stderr.log": "e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855", "check.stdout.log": "af8e9a371091dd2327af3784f48f79d137de7e7e00403d323475183ee8867f36", "command.txt": "423866794d3f8858d9a6bb858412ade9c602dd8319a3f5fa3bb4f296325d995b", "environment.json": "162b563ecf0da84fd1101dc77f607dafb1eec2142039cb33db0c21ef887ecdb0", "launch.json": "21cd754de1c2f026cf697711a92ee24c4a645c353e286beb534f24f3d44a0f2a", "manifest.yaml": "d6278bfda20ff82d9eaf8deceafd9dee3e391987aff7965720ebc07dd639b8e3", "raw-result.json": "f906d4d05bb1ff8128a64a56892fb6067ba89bac33f12338a2012e69013adefb", "stderr.log": "e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855", "stdout.log": "4914f20c5b5e938b1b0913279de8f604da79fd9dcb5a35827f4e994cb21def48"}, "authority": {"commit": "1e4fd5c77cf9f4ef7337ad0a9f29568174538a4b", "dispatch_plan_sha256": "73575df19302de2c6d2669300f09b59542ebcaad73f6d2944db8504b4f35b124", "epoch": 1, "expires_at": "2026-10-14T16:13:16+00:00", "owner": "lattice-executor-20261014"}, "check_returncode": 0, "detail": null, "finished_at": "2026-10-14T15:16:41.276773+00:00", "plan_sha256": "173e178ec736e4d59284bac8b5385176089e9aae9770d46aea08567704bf7c5f", "published": false, "returncode": 0, "run_id": "RUN-ENDO-910c24", "schema": "crypto.autoresearch.trial_receipt.v1", "scientific_conclusion": null, "started_at": "2026-10-14T15:16:41.067366+00:00", "status": "output_validated", "trial_id": "exact-pilot-seed1729"}

Control scope and graph

One retained supplied pool. Scoring and map-acquisition costs are incomplete; this is not a speedup comparison. All plotted values come directly from raw-result.json; no new sampling is performed. Uncertainty bars are inapplicable to these deterministic exact values. Plot source and extracted data are retained.

Limitations and missing data

The actual ambient order and candidate generators are supplied, not discovered. Selection counts omit isogeny construction/evaluation and are not performance claims. Full additive rank is not full order saturation; scalar and redundant maps are retained controls. Torsion division is certified only on the actual faithful supplied lattice. No finite-field characteristic-prime division, subgroup solve, or ECDLP consequence was measured. The integrity checker replays the implementation; it is not independent research review. No ECDLP canonical graph or frontier was changed.

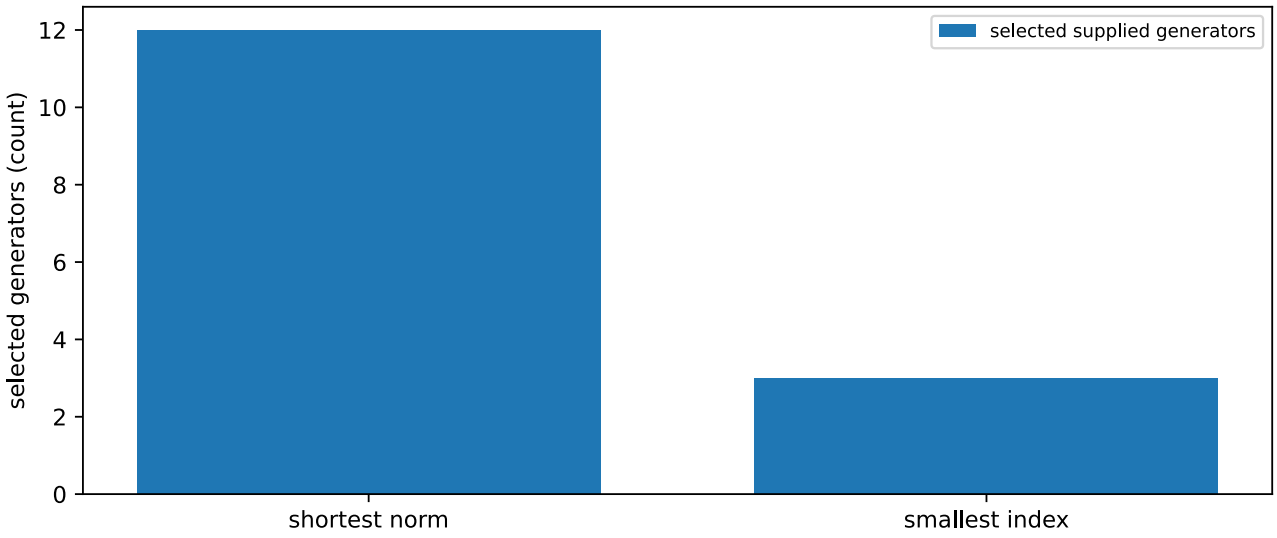

Figure 1: One retained supplied pool. Scoring and map-acquisition costs are incomplete; this is not a speedup comparison. Run RUN-ENDO-910c24.